

Optimal Investment Strategies with a Heath-Jarrow-Morton Term Structure of Interest Rates

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Claus Munk^{*†}

Department of Management

Odense University

Denmark

E-mail: `cmu@sam.sdu.dk`

Carsten Sørensen[†]

Department of Finance

Copenhagen Business School

Denmark

E-mail: `cso.fi@cbs.dk`

^{*}Corresponding author. Full address: Department of Management, Odense University, Campusvej 55, DK-5230 Odense M, Denmark. Phone: +45 6550 3257. Fax: +45 6593 0726.

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Abstract. We study the consumption and investment choice of a price-taking utility-maximizing investor having access to trade in stocks and interest rate dependent assets with a stochastically evolving term structure of interest rates. We derive explicit expressions for the optimal investment strategy of a HARA utility investor in a market where the term structure of interest rates is given by a general, possibly non-Markovian multi-factor Gaussian Heath-Jarrow-Morton term structure model and market prices of risk are deterministic. The optimal investment strategy combines the growth-optimal strategy and a single bond hedging the multi-factor interest rate risk. With utility from terminal wealth only, the hedge bond is the zero-coupon bond maturing at the horizon of the investor. With utility from intermediate consumption, the hedge bond has a continuous coupon proportional to the expected future consumption rate under the forward martingale measure.

Keywords. Asset allocation, martingale approach, stochastic investment opportunity set, non-Markov Gaussian HJM models, hedging

1 Introduction

The intertemporal consumption and investment decision of a utility-maximizing investor is a classical problem of financial economics. In two pathbreaking papers, Merton (1969, 1971) studies this problem in a continuous-time framework using dynamic programming. He derives explicit optimal consumption and investment strategies for an investor with an additively separable CRRA (Constant Relative Risk Aversion) utility from consumption and terminal wealth in a market with a constant investment opportunity set reflected by a constant short-term interest rate and lognormal financial asset prices. Merton (1973b) addresses the optimization problem under a stochastic investment opportunity set where the drift and diffusion coefficients of the asset prices depend on the current short-term interest rate, this rate follows a Markov process, and the market is complete. He shows that the optimal investment strategy combines the myopic, mean-variance portfolio and some portfolio hedging changes in the investment opportunity set, but he does not derive an explicit expression for the hedge portfolio.

In this paper we consider a complete financial market consisting of stocks and term-structure derivatives where the term structure of interest rates evolves stochastically according to a multi-factor Gaussian HJM model, cf. Heath, Jarrow, and Morton (1992). Furthermore, the market prices of risk related to both classes of assets are deterministic. Using an HJM framework allows for perfect calibration to the current term structure of interest rates. The short-term interest rate in (even Gaussian) HJM models is generally non-Markovian and, therefore, we cannot apply the traditional dynamic programming approach. Instead, we shall work with the martingale approach developed by Cox and Huang (1989) and Karatzas, Lehoczky, and Shreve (1987). With this approach the dynamic optimization problem is formulated as a static problem that can be solved in terms of the optimal consumption strategy and the optimal terminal wealth with standard Lagrangian theory, but the martingale approach does not provide an explicit expression for the optimal investment strategy.

We provide a characterization of the optimal investment strategy of a HARA (Hyperbolic Absolute Risk Aversion) investor in a general, non-Markov market setting and use this characterization to derive an explicit expression for the optimal investment strategy for a HARA investor in the multi-factor Gaussian HJM set-up. The optimal investment strategy combines the myopic, growth-optimal strategy and a single bond hedging changes in the investment opportunity set even though these changes are generated by a multi-dimensional Brownian Motion. With utility from terminal wealth only, the hedge bond is the zero-coupon bond maturing at the horizon of the investor. With utility from intermediate consumption, the hedge bond has a continuous coupon proportional to the expected future consumption rate under the forward martingale measure. We first demon-

strate these results for a CRRA utility investor and then generalize the results to the case where the investor has a power utility for consumption or wealth in excess of some subsistence level, the economically most appealing type of utility function in the HARA class (see the discussion in Section 3).

In the literature there has recently been a number of studies of optimal investment strategies in dynamic markets with specific assumptions on the changes in the investment opportunity set. Kim and Omberg (1996) study a market where the interest rate is constant and the Sharpe ratio of the only available risky asset follows an Ornstein-Uhlenbeck process imperfectly correlated with the risky asset price itself. In this incomplete market setting, they derive an explicit expression for the optimal portfolio process for an investor with HARA utility from terminal wealth and study deviations from the myopic, growth-optimal portfolio. Chacko and Viceira (1999) focus on the case where the volatility of the single risky asset price follows a Markov process imperfectly correlated with the price process implying incompleteness of the market. Furthermore, the short-term interest rate is assumed constant and the expected return on the risky asset is either constant or an affine function of the instantaneous variance (the volatility squared). For an infinite horizon investor with a recursive utility function, they derive a closed-form approximate solution for the optimal investment strategy and discuss the deviations from the myopic strategy.

Detemple, Garcia, and Rindisbacher (1999) consider a complete market where changes in the investment opportunity set are driven by a multi-dimensional state variable following a Markov diffusion. For an investor maximizing utility of terminal wealth, they are able to express the optimal investment strategy as a combination of the myopic, growth-optimal strategy and two terms representing the hedge against changes in the short-term interest rate and the market prices of risk, respectively. The two hedge components are given as conditional expectations involving integrals of Malliavin derivatives with respect to the short-term interest rate and the market price of risk. In the special setting of a CRRA investor in a complete market with a single risky asset where the short-term interest rate and the market price of risk follow two particular diffusion processes that are both perfectly correlated with the risky asset price, they are able to derive the hedge strategies in closed-form. For other settings they demonstrate how to compute the hedge components numerically by simulations using their Malliavin formulation.

Brennan, Schwartz, and Lagnado (1997) solve numerically for the optimal investment strategy of an investor with CRRA utility from terminal wealth in a market with stocks and bonds driven by three variables: the short-term and the long-term interest rates and a stochastic dividend yield on the stock. They investigate how the optimal investment strategy deviates from the myopic strategy for different investment horizons.

Finally, Sørensen (1999) considers an investor with CRRA utility from terminal wealth in

a complete market setting close to ours. In the market a single stock with a constant excess expected return and volatility is traded and the short-term interest rates and the prices of bonds are generated by the one-factor Vasicek (1977) model. The shocks to the stock price and to the short rate are imperfectly correlated. He shows that the optimal investment strategy combines the myopic, growth-optimal portfolio and the zero-coupon bond maturing at the investment horizon and that the relative weights of the two components only depend on the risk aversion parameter of the investor.

In contrast to all the papers referred above, we allow for non-Markovian changes in the opportunity set in modeling the term structure dynamics by a general multi-factor Gaussian HJM model, but we are still able to derive an explicit expression for the optimal investment strategy of investors with HARA utility of terminal wealth and/or of continuous consumption. The optimal strategy has the same structure as in the simpler one-factor Markovian setting of Sørensen. It combines the growth-optimal strategy and a single bond hedging the multi-factor interest rate risk.

The rest of the paper is organized as follows. In Section 2 we set up a general complete financial market and review the martingale approach for general time-separable utility functions. In Section 3 we focus on CRRA and HARA utility and derive a general condition that the portfolio optimally hedging changes in the opportunity set must satisfy. The financial model is specialized in Section 4 to a market where interest rates are generated by a multi-factor Gaussian HJM model and market prices of risk are deterministic functions of time. In Section 5 we use the general hedging condition to derive an explicit expression for the optimal investment strategy for an investor with CRRA or HARA utility from terminal wealth only in the specialized market setting. A similar analysis for the more complicated case with utility from intermediate consumption as well as terminal wealth is given in Section 6. Finally, Section 7 concludes the paper.

2 A general complete market consumption-investment problem

The martingale approach was suggested by Karatzas, Lehoczky, and Shreve (1987) and Cox and Huang (1989, 1991) as an alternative to the standard dynamic programming approach introduced by Merton (1969). Below, we shall review the main aspects of the approach and focus on the results relevant for the remaining sections of the paper. For a comprehensive overview of the approach, the interested reader is referred to Karatzas and Shreve (1998, Ch. 3) or Korn (1997, Ch. 3).

2.1 The model

Information structure Let w be a d -dimensional Wiener process on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$, where $\mathbb{F} = \{\mathcal{F}_t \mid t \in \mathcal{T}\}$ is a right-continuous filtration. Here we take $\mathcal{T} = [0, T]$ for

some $T > 0$ representing the time horizon of the investor considered. $\mathcal{F}_t$ is the augmentation of the σ -algebra generated by $\{w(u) \mid 0 \leq u \leq t\}$. It is assumed that $\mathcal{F}_0$ is the σ -algebra generated by the zero sets of P and that $\mathcal{F} = \mathcal{F}_T$. Below, all statements involving stochastic variables are assumed to hold *almost surely* wrt. $\mathbb{P}$ and all stochastic processes are assumed to be adapted to $\mathbb{F}$.

Financial assets The agents in the economy have access to continuous trading in an instantaneously riskless asset called *the savings account* with price $A(t)$ satisfying

$$dA(t) = r(t)A(t) dt \quad (1)$$

where r is the continuously compounded short-term interest rate process, and d risky assets with prices given by the vector $P(t) = (P_1(t), \dots, P_d(t))^\top$ satisfying

$$dP(t) = \text{diag}(P(t)) [(r(t)\mathbf{1} + \sigma(t)\lambda(t)) dt + \sigma(t) dw(t)] . \quad (2)$$

Here, λ is an $\mathbb{R}^d$ -valued $\mathcal{L}^2[0, T]$ stochastic process of market prices of risk, and σ is an $\mathbb{R}^{d \times d}$ -valued stochastic process of volatilities.¹ The processes r , λ , and σ are assumed to be progressively measurable with respect to $\mathbb{F}$ and such that the equations (1) and (2) are well-defined.² Note that we allow for non-Markovian dynamics of the investment opportunity set. The volatility process $\sigma(t)$ is assumed to satisfy the non-degeneracy assumption

$$\exists \epsilon > 0 \forall (x, t) \in \mathbb{R}^d \times \mathcal{T} : x^\top \sigma(t) \sigma(t)^\top x \geq \epsilon \|x\|^2. \quad (3)$$

As a consequence of condition (3), σ has full rank d implying the dynamic completeness of the market.³ Define the processes

$$\begin{aligned} Z_{t_0}(t) &= \exp \left\{ - \int_{t_0}^t \lambda(s)^\top dw(s) - \frac{1}{2} \int_{t_0}^t \|\lambda(s)\|^2 ds \right\}, \quad t \geq t_0, \\ \zeta_{t_0}(t) &= \exp \left\{ - \int_{t_0}^t r(s) ds - \int_{t_0}^t \lambda(s)^\top dw(s) - \frac{1}{2} \int_{t_0}^t \|\lambda(s)\|^2 ds \right\}, \quad t \geq t_0. \end{aligned} \quad (4)$$

We assume that Z_{t_0} is a martingale, which, e.g., is the case if λ satisfies the Novikov condition. The process ζ_{t_0} is the unique state-price density⁴, and $d\mathbb{Q}/d\mathbb{P} = Z_{t_0}(T)$ defines the unique equivalent spot martingale measure $\mathbb{Q}$ given the information available at time t_0 .

¹ $\mathcal{L}^2[0, T]$ is the set of adapted stochastic processes z such that $\int_0^T \|z(t)\|^2 dt < \infty$ almost surely. Similarly, $\mathcal{L}^1[0, T]$ is the set of adapted processes z with $\int_0^T \|z(t)\| dt < \infty$ almost surely.

² In addition to the $\mathcal{L}^2[0, T]$ assumption on λ , it suffices that $r \in \mathcal{L}^1[0, T]$ and $\sigma \in \mathcal{L}^2[0, T]$.

³ The market is potentially complete, but by restricting the set of admissible portfolio processes, various types of incompleteness can be modeled. This complicates the solution of the utility maximization problem considerably, cf. Cvitanic and Karatzas (1992) and Cuoco (1997).

⁴ This is sometimes called the *state-price deflator* or the *pricing kernel*.

Optimization problem We consider an investor seeking to maximize his expected remaining life-time utility by choosing a trading and consumption strategy appropriately. If we let $c(t)$ denote the consumption rate at time t and X the time T terminal wealth, then the expected utility of the agent over the time interval $[t_0, T]$ is

$$\mathbb{E}_{t_0} \left[\int_{t_0}^T U_1(c(t), t) dt + U_2(X, T) \right],$$

where $\mathbb{E}_{t_0}[\cdot]$ is short for $\mathbb{E}[\cdot | \mathcal{F}_{t_0}]$. Here, $U_1(\cdot, t)$ is defined over an interval $[\underline{c}(t), \infty)$ with $\underline{c}(t) \geq 0$ representing the time t subsistence consumption level. We assume that $U_1(\cdot, t)$ is a strictly increasing and concave $C^1(\underline{c}(t), \infty)$ function with $U_1'(\infty, t) \equiv \lim_{c \uparrow \infty} U_1'(c, t) = 0$ and $U_1'(\underline{c}(t), t) \equiv \lim_{c \downarrow \underline{c}(t)} U_1'(c, t) = \infty$ where the primes denote partial derivatives with respect to the first argument. Similarly, $U_2(\cdot, T)$ is defined over an interval $[\underline{X}, \infty)$ with $\underline{X} \geq 0$ denoting subsistence terminal wealth. Also $U_2(\cdot, T)$ is assumed to be a strictly increasing and concave $C^1(\underline{X}, \infty)$ function with $U_2'(\infty, T) \equiv \lim_{X \uparrow \infty} U_2'(X, T) = 0$ and $U_2'(\underline{X}, T) \equiv \lim_{X \downarrow \underline{X}} U_2'(X, T) = \infty$.

Since we assume that the investor does not receive income from non-traded assets, zero is a natural lower bound on the wealth process.⁵ Therefore, we can represent the trading strategy of the investor by an $\mathbb{R}^d$ -valued progressively measurable stochastic process $\pi = (\pi_1, \dots, \pi_d)^\top$ with $\pi_i(t)$ denoting the fraction of wealth invested in the i 'th risky asset at time t . The fraction of wealth invested in the locally riskless asset is residually determined as $\pi_0(t) = 1 - \pi(t)^\top \mathbf{1}$. A consumption strategy over the interval $[t_0, T]$ is a progressively measurable process c with $c(t) \geq \underline{c}(t)$, $t \in [t_0, T]$.

Given a trading strategy π and a consumption strategy c , the wealth $X_{x, t_0}^{\pi, c}$ of the investor evolves according to

$$dX_{x, t_0}^{\pi, c}(t) = [r(t)X_{x, t_0}^{\pi, c}(t) + X_{x, t_0}^{\pi, c}(t)\pi(t)^\top \sigma(t)\lambda(t) - c(t)] dt + X_{x, t_0}^{\pi, c}(t)\pi(t)^\top \sigma(t) dw(t) \quad (5)$$

with $X_{x, t_0}^{\pi, c}(t_0) = x$. We denote by $\mathcal{C}_{t_0}$ the set of all consumption processes $c \in \mathcal{L}^1[t_0, T]$ with $c(t) \geq \underline{c}(t)$ for all $t \in [t_0, T]$ and by $\mathcal{L}_+^1$ the set of $\mathcal{F}_T$ -measurable random variables X with finite expectations and $X \geq \underline{X}$. A consumption/terminal wealth pair $(c, X) \in \mathcal{C}_{t_0} \times \mathcal{L}_+^1$ is called admissible over $[t_0, T]$ with wealth x at time t_0 , if a trading strategy $\pi \in \mathcal{L}^2[t_0, T]$ exists such that $X_{x, t_0}^{\pi, c}(T) = X$. In that case the trading strategy π is said to finance (c, X) . The set of such admissible pairs (c, X) is denoted by $\mathcal{A}(x, t_0)$.

The utility maximization problem of the investor is then stated as follows:

$$V(x, t_0) = \sup_{(c, X) \in \mathcal{A}(x, t_0)} \mathbb{E}_{t_0} \left[\int_{t_0}^T U_1(c(s), s) ds + U_2(X, T) \right]. \quad (6)$$

The value function V is assumed to be finite for all (x, t_0) .

⁵ Allowing for a stochastic income stream is straightforward if the income process is spanned by traded assets. The effect of undiversifiable income risk is studied, e.g., by Duffie, Fleming, Soner, and Zariphopoulou (1997) and Munk (1999).

2.2 The martingale approach

The basic idea of the martingale approach is to focus on the *static* optimization problem

$$\begin{aligned} \sup_{(c,X) \in \mathcal{C} \times \mathcal{L}_+^1} & \mathbb{E}_{t_0} \left[\int_{t_0}^T U_1(c(t), t) dt + U_2(X, T) \right], \\ \text{s.t. } & \mathbb{E}_{t_0} \left[\int_{t_0}^T \zeta(t)c(t) dt + \zeta(T)X \right] \leq x. \end{aligned} \quad (7)$$

instead of the *dynamic* problem (5)–(6). Note that there will be no feasible solution to (7) if $x < L(t_0)$, where

$$L(t_0) = \mathbb{E}_{t_0} \left[\int_{t_0}^T \zeta_{t_0}(t)\underline{c}(t) dt + \zeta_{t_0}(T)\underline{X} \right] \quad (8)$$

is the minimum wealth required to finance the subsistence consumption and terminal wealth. In the following we shall assume that $x > L(t_0)$. Due to the assumption of infinite marginal utility at the subsistence level, we can ignore the constraints $c(t) \geq \underline{c}(t)$ and $X \geq \underline{X}$ when maximizing.

By Lagrangian theory the optimal solution to the static problem must satisfy

$$U'_1(c(t), t) = \psi \zeta_{t_0}(t), \quad U'_2(X, T) = \psi \zeta_{t_0}(T),$$

where ψ is such that the inequality constraint holds as an equality. Let $I_1(\cdot, t)$ denote the inverse of the marginal utility function $U'_1(\cdot, t)$ and $I_2(\cdot, T)$ the inverse of $U'_2(\cdot, T)$. Define

$$\mathcal{H}_{t_0}(\psi) = \mathbb{E}_{t_0} \left[\int_{t_0}^T \zeta_{t_0}(t) I_1(\psi \zeta_{t_0}(t), t) dt + \zeta_{t_0}(T) I_2(\psi \zeta_{t_0}(T), T) \right].$$

Obviously, $\mathcal{H}_{t_0}(\cdot)$ is a decreasing function. Assume that $\mathcal{H}_{t_0}(\psi)$ is finite for all ψ . Then $\mathcal{H}_{t_0}(\cdot)$ has an inverse denoted by $\mathcal{Y}_{t_0}(\cdot)$ so that $\psi = \mathcal{Y}_{t_0}(x)$ and the optimal solution to (7) can be written as

$$c^*(t) = I_1(\mathcal{Y}_{t_0}(x) \zeta_{t_0}(t), t), \quad t \in [t_0, T], \quad (9)$$

$$X^* = I_2(\mathcal{Y}_{t_0}(x) \zeta_{t_0}(T), T). \quad (10)$$

One of the main results of the martingale approach is that (9)–(10) is also the optimal solution to the dynamic problem (6). The wealth process under the optimal policy is given by

$$X_{x,t_0}^*(t) = \frac{1}{\zeta_{t_0}(t)} \mathbb{E}_t \left[\int_t^T \zeta_{t_0}(s) c^*(s) ds + \zeta_{t_0}(T) X^* \right]. \quad (11)$$

The value function can be written as

$$\begin{aligned} V(x, t_0) &= \mathbb{E}_{t_0} \left[\int_{t_0}^T U_1(c^*(s), s) ds + U_2(X^*, T) \right] \\ &= \mathbb{E}_{t_0} \left[\int_{t_0}^T U_1(I_1(\mathcal{Y}_{t_0}(x) \zeta_{t_0}(t), t), t) dt + U_2(I_2(\mathcal{Y}_{t_0}(x) \zeta_{t_0}(T), T), T) \right]. \end{aligned}$$

A drawback of the martingale approach is that the optimal investment policy is only given implicitly by the martingale representation theorem. Therefore, it is generally not clear how to implement (c^*, X^*) by a trading strategy π^* such that

$$X_{x,t_0}^{\pi^*, c^*}(t) = X_{x,t_0}^*(t), \quad t \in [t_0, T].$$

In the case of logarithmic utility, it can be shown (see, e.g., Karatzas and Shreve (1998, Example 3.6.6)) that the optimal investment strategy is to invest the fractions

$$\pi_G(t) = (\sigma(t)^\top)^{-1} \lambda(t) \quad (12)$$

of wealth in the d risky assets and the remaining fraction $1 - \pi_G(t)^\top \mathbf{1}$ in the savings account. We will refer to the portfolio $(\pi_G(t), 1 - \pi_G(t)^\top \mathbf{1})$ as the growth-optimal investment strategy.

For general utility functions, the optimal investment strategy can be represented rather abstractly in terms of stochastic integrals of Malliavin derivatives by the Clark-Ocone formula, cf. Ocone and Karatzas (1991), but to derive an explicit expression for the optimal portfolio for non-logarithmic utility functions it is generally recognized that the price dynamics must be specialized. Cox and Huang (1989) show that when the state-price density and the risky asset prices constitute a Markovian system, the optimal investment strategy is given in terms of the solution of a linear second order partial differential equation. More explicit results are given in the case of a deterministically changing investment opportunity set, cf., e.g., Cvitanić and Karatzas (1992). In the following sections we, on the other hand, provide a closed-form expression for the optimal investment strategy in a non-Markov market setting.

3 Results for HARA utility in a general market setting

We will characterize the optimal investment strategy of a HARA utility investor in the general market setting outlined above. Let us first consider the case of CRRA utility:

$$U_1(c, t) = e^{-\beta(t-t_0)} \frac{c^{1-\gamma}}{1-\gamma}, \quad U_2(X, T) = e^{-\beta(T-t_0)} \frac{X^{1-\gamma}}{1-\gamma}, \quad \gamma > 0, \quad (13)$$

which for $\gamma = 1$ is interpreted as the limiting case of logarithmic utility. Straightforward computations give $\mathcal{H}_{t_0}(\psi) = Q(t_0)\psi^{-1/\gamma}$ and $\mathcal{Y}_{t_0}(x) = x^{-\gamma}Q(t_0)^\gamma$, where

$$Q(t) = \int_t^T e^{-\frac{\beta}{\gamma}(s-t)} \mathbb{E}_t \left[\zeta_t(s)^{\frac{\gamma-1}{\gamma}} \right] ds + e^{-\frac{\beta}{\gamma}(T-t)} \mathbb{E}_t \left[\zeta_t(T)^{\frac{\gamma-1}{\gamma}} \right]. \quad (14)$$

The optimal policies are

$$c^*(t) = e^{-\frac{\beta}{\gamma}(t-t_0)} (\mathcal{Y}_{t_0}(x)\zeta_{t_0}(t))^{-1/\gamma} = e^{-\frac{\beta}{\gamma}(t-t_0)} \zeta_{t_0}(t)^{-1/\gamma} Q(t_0)^{-1} x, \quad (15)$$

$$X^* = e^{-\frac{\beta}{\gamma}(T-t_0)} (\mathcal{Y}_{t_0}(x)\zeta_{t_0}(T))^{-1/\gamma} = e^{-\frac{\beta}{\gamma}(T-t_0)} \zeta_{t_0}(T)^{-1/\gamma} Q(t_0)^{-1} x, \quad (16)$$

and the value function is

$$V(x, t_0) = \frac{Q(t_0)^\gamma x^{1-\gamma}}{1-\gamma}. \quad (17)$$

Substituting (15) and (16) into (11) we get that the optimally invested wealth at time t can be written as

$$X_{x,t_0}^*(t) = x e^{-\frac{\beta}{\gamma}(t-t_0)} \zeta_{t_0}(t)^{-1/\gamma} \frac{Q(t)}{Q(t_0)}. \quad (18)$$

As a consequence the optimal consumption strategy can, even in this general setting, be written in a feedback-type form as

$$c^*(t) = \frac{X_{x,t_0}^*(t)}{Q(t)}.$$

The expression (18) allows a general characterization of the optimal investment strategy. First, note that $Q(t)$ is a positive, adapted stochastic process so that we can write the dynamics of $Q(t)$ as

$$dQ(t) = Q(t) [\mu_Q(t) dt + \sigma_Q(t)^\top dw(t)]$$

for some drift process $\mu_Q(t)$ and some volatility process $\sigma_Q(t)$. Also note that from (4) the dynamics of the state-price density is given by

$$d\zeta_{t_0}(t) = -\zeta_{t_0}(t) [r(t) dt + \lambda(t)^\top dw(t)].$$

The optimally invested wealth is given in (18) as a function of time, $\zeta_{t_0}(t)$, and $Q(t)$, so an application of Itô's Lemma yields that the dynamics of the optimally invested wealth can be written as

$$\begin{aligned} dX_{x,t_0}^*(t) = X_{x,t_0}^*(t) & \left[\left(-\frac{\beta}{\gamma} + \frac{r(t)}{\gamma} + \mu_Q(t) + \frac{1+\gamma}{2\gamma^2} \|\lambda(t)\|^2 - \frac{1}{\gamma} \lambda(t)^\top \sigma_Q(t) \right) dt \right. \\ & \left. + \frac{1}{\gamma} \lambda(t)^\top dw(t) + \sigma_Q(t)^\top dw(t) \right], \quad t_0 \leq t \leq T. \end{aligned} \quad (19)$$

The optimal investment strategy of a CRRA investor with a constant investment opportunity set is to invest the fractions $\pi_G(t)/\gamma$ of wealth in the d risky assets and the fraction $1 - \pi_G(t)^\top \mathbf{1}/\gamma$ of wealth in the savings account, where $\pi_G(t)$ is given by (12), cf. Merton (1969). If we decompose the investment strategy of the investor as

$$\begin{pmatrix} \pi(t) \\ \pi_0(t) \end{pmatrix} = \begin{pmatrix} \frac{1}{\gamma} \pi_G(t) \\ 1 - \pi_G(t)^\top \mathbf{1}/\gamma \end{pmatrix} + \begin{pmatrix} \pi^H(t) \\ 1 - \pi^H(t)^\top \mathbf{1} \end{pmatrix}, \quad (20)$$

we can interpret $\pi^H(t)$ as the portfolio hedging against changes in the investment opportunity set. With a given hedge strategy $\pi^H(t)$ and a consumption strategy $c(t) = X_{x,t_0}^{\pi^H}(t)/Q(t)$, the wealth

of the investor $X_{x,t_0}^{\pi^H}(t)$ evolves according to

$$dX_{x,t_0}^{\pi^H}(t) = X_{x,t_0}^{\pi^H}(t) \left[\left(r(t) - \frac{1}{Q(t)} + \frac{1}{\gamma} \|\lambda(t)\|^2 + \pi^H(t)^\top \sigma(t) \lambda(t) \right) dt + \frac{1}{\gamma} \lambda(t)^\top dw(t) + \pi^H(t)^\top \sigma(t) dw(t) \right], \quad t_0 \leq t \leq T.$$

In order to replicate the dynamics of the optimally invested wealth in (19) the diffusion coefficients must be matched (almost surely). Hence, the hedge portfolio $\pi^H(t)$ is to be chosen such that

$$\pi^H(t)^\top \sigma(t) = \sigma_Q(t)^\top, \quad (21)$$

i.e. such that the volatility of the value of the hedge portfolio is equal to the volatility of the process $Q(t)$. In the following sections we will consider a specialized, but still non-Markov, market setting where $\sigma_Q(t)$ can be explicitly computed and an explicit expression for the hedge portfolio can be derived.

Next we will allow more general utility functions in the HARA class than the CRRA function. As discussed by Kim and Omberg (1996), the HARA class of utility functions $U(X)$ can be divided into three subclasses:

1. $U(X) = (X - \underline{X})^{1-\gamma}/(1-\gamma)$ with $\gamma > 0$ (interpreted as $U(X) = \ln(X - \underline{X})$ for $\gamma = 1$) defined for all $X > \underline{X}$,
2. $U(X) = -(\underline{X} - X)^{1-\gamma}$ with $\gamma < 0$ defined for all $X < \underline{X}$,
3. $U(X) = -\exp(-aX)$ with $a > 0$ defined for all $X \in \mathbb{R}$.

We will only consider cases where the obtainable wealth and consumption levels are bounded below by a non-negative constant and unbounded from above and will therefore focus on the first subclass of functions (with $\underline{X} \geq 0$). This is also the only subclass of HARA functions satisfying the regularity conditions required for the application of the martingale method.

Therefore consider an investor with utility functions

$$U_1(c, t) = e^{-\beta(t-t_0)} \frac{(c - \underline{c}(t))^{1-\gamma}}{1-\gamma}, \quad U_2(X, T) = e^{-\beta(T-t_0)} \frac{(X - \underline{X})^{1-\gamma}}{1-\gamma}, \quad (22)$$

where $\underline{c}(t)$ and $\underline{X}$ are non-negative real numbers that can be interpreted as the subsistence time t consumption and terminal wealth level, respectively. Bardhan (1994) has shown that the optimal policy is to invest $L(t_0)$ defined in (8) in order to replicate the subsistence lower bound and allocate the remaining $x - L(t_0)$ as if the subsistence level was not present. The optimal policies are, hence,

$$\begin{aligned} c^*(t) &= e^{-\frac{\beta}{\gamma}(t-t_0)} \zeta_{t_0}(t)^{-1/\gamma} Q(t_0)^{-1} (x - L(t_0)) + \underline{c}(t), \\ X^* &= e^{-\frac{\beta}{\gamma}(T-t_0)} \zeta_{t_0}(T)^{-1/\gamma} Q(t_0)^{-1} (x - L(t_0)) + \underline{X}. \end{aligned} \quad (23)$$

The value function becomes

$$V(x, t_0) = \frac{Q(t_0)^\gamma (x - L(t_0))^{1-\gamma}}{1-\gamma}.$$

Since

$$X_{x,t_0}^*(t) = e^{-\frac{\beta}{\gamma}(t-t_0)} \zeta_{t_0}(t)^{-1/\gamma} (x - L(t_0)) \frac{Q(t)}{Q(t_0)} + L(t),$$

the time t optimal consumption rate can be written in terms of time t optimal wealth as

$$c^*(t) = \frac{X_{x,t_0}^*(t) - L(t)}{Q(t)} + \underline{c}(t).$$

In the remaining sections, we will apply the results above in order to derive the optimal investment strategies of a HARA investor in a specialized market setting.

4 A market with a Gaussian HJM term structure and deterministic market prices of risk

We will now specialize the general set-up of the previous section. The financial market is separated into a stock market and a bond market. We write the d -dimensional Wiener process generating the information structure of the financial market as $w = (w_B, w_S)$, where w_B is of dimension $k < d$ and w_S of dimension $l = d - k$. We assume that the dynamics of the term structure of interest rates, and hence the dynamics of prices on bonds and other term structure derivatives traded at the bond market, is affected only by w_B . The dynamics of the prices of the stocks may depend on both w_B and w_S which allows for correlation between stocks and term structure derivatives. More specifically, the investor can trade in the instantaneously riskless savings account, in k term-structure derivatives with price dynamics given by

$$dB_t = \text{diag}(B_t) [(r(t)\mathbf{1}_k + \sigma_B(t)\lambda_B(t)) dt + \sigma_B(t) dw_B(t)], \quad (24)$$

and in l stocks with price dynamics

$$dS_t = \text{diag}(S_t) [(r(t)\mathbf{1}_l + \varphi_S(t)) dt + \sigma_{S1}(t) dw_B(t) + \sigma_{S2}(t) dw_S(t)].$$

Here $\sigma_B(t)$, $\sigma_{S1}(t)$ and $\sigma_{S2}(t)$ are matrix valued processes of dimension $k \times k$, $l \times k$ and $l \times l$, respectively, with $\sigma_B(t)$ and $\sigma_{S2}(t)$ non-singular so that the market is complete. Changes in the returns of the term structure derivatives and the stocks are correlated with $k \times l$ covariance matrix $\sigma_B(t)\sigma_{S1}(t)^\top$. In this market the relative risk process λ (which is not dependent on the particular set of assets chosen) is

$$\lambda(t) = \begin{pmatrix} \lambda_B(t) \\ \lambda_S(t) \end{pmatrix}$$

where

$$\lambda_S(t) = \sigma_{S2}(t)^{-1} \varphi_S(t) - \sigma_{S2}(t)^{-1} \sigma_{S1}(t) \lambda_B(t).$$

More specifically, we assume that the dynamics of the term structure of interest rates can be described by a k -factor model of the HJM-class introduced by Heath, Jarrow, and Morton (1992). An HJM-framework is natural since it allows for perfect calibration of the model to existing market prices. For any τ the dynamics of the τ -maturity instantaneous forward rate is

$$f(t, \tau) = f(t_0, \tau) + \int_{t_0}^t \alpha(s, \tau) ds + \int_{t_0}^t \sigma_f(s, \tau)^\top dw_B(s), \quad (25)$$

where $\sigma_f(\cdot, \tau)$ is an $\mathbb{R}^k$ -valued process. Due to the no-arbitrage drift restriction, we have that

$$\alpha(t, \tau) = \sigma_f(t, \tau)^\top \left(\lambda_B(t) + \int_t^\tau \sigma_f(t, u) du \right).$$

The short rate is given by $r(t) = f(t, t)$ and evolves, therefore, according to the equation

$$r(t) = f(t_0, t) + \int_{t_0}^t \alpha(s, t) ds + \int_{t_0}^t \sigma_f(s, t)^\top dw_B(s). \quad (26)$$

Among the many term-structure derivatives, we focus on default-free bonds. The dynamics of the price $P(t, \tau) = \exp(-\int_t^\tau f(t, s) ds)$ of the zero-coupon bond maturing at time τ is given by

$$dP(t, \tau) = P(t, \tau) \left[(r(t) + \sigma_P(t, \tau)^\top \lambda_B(t)) dt + \sigma_P(t, \tau)^\top dw_B(t) \right], \quad (27)$$

where $\sigma_P(t, \tau) = -\int_t^\tau \sigma_f(t, u) du$. For later use we will also consider a bond paying a continuous coupon of $k(t)$ up to time T and a lump sum payment of K at time T . The time t price of such a bond is

$$P^{\text{cpn}}(t) = \int_t^T k(s) P(t, s) ds + K P(t, T).$$

Applying the Leibnitz-type rule for stochastic processes stated in Lemma 1 in the Appendix, we see that the coupon bond price evolves as

$$dP^{\text{cpn}}(t) = -k(t) dt + P^{\text{cpn}}(t) \left[(r(t) + \sigma_{P^{\text{cpn}}}(t)^\top \lambda_B(t)) dt + \sigma_{P^{\text{cpn}}}(t)^\top dw_B(t) \right],$$

where

$$\sigma_{P^{\text{cpn}}}(t) = \frac{\int_t^T k(s) P(t, s) \sigma_P(t, s) ds + K P(t, T) \sigma_P(t, T)}{\int_t^T k(s) P(t, s) ds + K P(t, T)}. \quad (28)$$

In order to obtain closed-form expressions for optimal investment strategies we shall employ the following assumption.

Assumption 1 *The relative risk process $\lambda(t)$ and the forward rate volatilities $\sigma_f(t, \tau)$ are deterministic functions of time.*

In particular, we limit ourselves to Gaussian models of the term structure. Despite obvious shortcomings, such as not ruling out negative interest rates, multi-factor Gaussian models are often used for derivative pricing since they allow closed-form solution for most European-type term structure contingent claims, cf., e.g., Amin and Jarrow (1992) and Brace and Musiela (1994). As we shall see in the following sections, the Gaussian assumption also allows closed-form expressions for optimal investment strategies. Recall that not even in Gaussian HJM-models is the short rate process necessarily Markovian. Only if $\sigma_f(t, \tau)$ can be separated as $\sigma_f(t, \tau) = G(t)H(\tau)$, where H is a real-valued continuously differentiable function that never changes sign and G is an $\mathbb{R}^k$ -valued continuously differentiable function, is the short rate Markovian, cf. Carverhill (1994). Also note that we do not assume that the diffusion coefficients σ_B , σ_{S1} , and σ_{S2} are deterministic.

From Assumption 1 it follows that the state-price density $\zeta_t(t)$ defined in (4) is lognormally distributed. Hence, it is possible to compute in closed-form the expectations $\mathbb{E}_t \left[\zeta_t(s)^{\frac{\gamma-1}{\gamma}} \right]$ that enter the $Q(\cdot)$ -function defined in (14). First note that

$$P(t, s) = \mathbb{E}_t [\zeta_t(s)] = \exp \left\{ \mathbb{E}_t [\ln \zeta_t(s)] + \frac{1}{2} \text{Var}_t [\ln \zeta_t(s)] \right\}.$$

It follows that

$$\begin{aligned} \mathbb{E}_t \left[\zeta_t(s)^{\frac{\gamma-1}{\gamma}} \right] &= \exp \left\{ \frac{\gamma-1}{\gamma} \mathbb{E}_t [\ln \zeta_t(s)] + \frac{(\gamma-1)^2}{2\gamma^2} \text{Var}_t [\ln \zeta_t(s)] \right\} \\ &= P(t, s)^{\frac{\gamma-1}{\gamma}} \exp \left\{ \frac{1-\gamma}{2\gamma^2} g(t, s) \right\}, \end{aligned} \quad (29)$$

where for notational simplicity we have introduced the deterministic function

$$g(t, s) = \text{Var}_t [\ln \zeta_t(s)],$$

which can be rewritten using (4) and (26) as

$$\begin{aligned} g(t, s) &= \text{Var}_t \left[- \int_t^s r(u) du - \frac{1}{2} \int_t^s \|\lambda(u)\|^2 du - \int_t^s \lambda(u)^\top dw(u) \right] \\ &= \int_t^s \|\lambda(u)\|^2 du + \int_t^s \left\| \int_u^s \sigma_f(u, \tau) d\tau \right\|^2 du + 2 \int_t^s \lambda_B(u)^\top \left[\int_u^s \sigma_f(u, \tau) d\tau \right] du \\ &= \int_t^s \|\lambda(u)\|^2 du + \int_t^s \|\sigma_P(u, s)\|^2 du - 2 \int_t^s \lambda_B(u)^\top \sigma_P(u, s) du. \end{aligned} \quad (30)$$

In the following sections, we shall employ this result to derive optimal investment strategies for CRRA and HARA utility functions.

5 Optimal investment strategies with utility from terminal wealth only

In this section we study the problem of an investor having utility from terminal wealth only and access to trade in the financial market outlined in Section 4. First, we consider the CRRA utility function $U_2(X, T)$ given in (13).

With no intermediate consumption, the process $Q(t)$ defined in (14) simplifies to

$$Q(t) = e^{-\frac{\beta}{\gamma}(T-t)} \mathbb{E}_t \left[\zeta_t(T)^{\frac{\gamma-1}{\gamma}} \right].$$

In the setting of Section 4 we can rewrite $Q(t)$ using (29) as

$$Q(t) = e^{-\frac{\beta}{\gamma}(T-t) + \frac{1-\gamma}{2\gamma^2}g(t,T)} P(t,T)^{\frac{\gamma-1}{\gamma}}. \quad (31)$$

Substituting this expression into (17), we see that the value function of this investor can be written as

$$V(x, t_0) = \frac{1}{1-\gamma} e^{-\beta(T-t_0)} \left(\frac{x}{P(t_0, T)} \right)^{1-\gamma} \exp \left\{ \frac{1-\gamma}{2\gamma} g(t_0, T) \right\}.$$

Note that $x/P(t_0, T)$ can be interpreted as the forward price of current wealth, which will be the terminal wealth of an agent investing his entire wealth x in zero-coupon bonds maturing at time T . This terminal wealth level is adjusted by the exponential term to reflect the benefits from dynamic trading strategies. These benefits depend on the risk aversion of the investor through the appearance of the relative risk aversion parameter γ .

We will find the optimal hedge portfolio by an application of the general characterization in (21) so that the optimal investment strategy follows from the portfolio decomposition in (20). To do that we must derive the dynamics of the process $Q(t)$ which is given by (31) as a function of time and the zero-coupon bond price $P(t, T)$. Noting that

$$\frac{\partial g(t, s)}{\partial t} = -\|\lambda(t)\|^2 - \|\sigma_P(t, s)\|^2 + 2\lambda_B(t)^\top \sigma_P(t, s), \quad (32)$$

Itô's Lemma, (27), and (30) imply that

$$dQ(t) = Q(t) \left[\left(\frac{\beta}{\gamma} + \frac{\gamma-1}{\gamma} r(t) - \frac{1-\gamma}{2\gamma^2} \|\lambda(t)\|^2 + \frac{(1-\gamma)^2}{\gamma^2} \lambda_B(t)^\top \sigma_P(t, T) \right) dt + \frac{\gamma-1}{\gamma} \sigma_P(t, T)^\top dw_B(t) \right]$$

corresponding to

$$\sigma_Q(t)^\top = \left(\frac{\gamma-1}{\gamma} \sigma_P(t, T)^\top, \mathbf{0}^\top \right).$$

From the general condition (21), the hedge portfolio $\pi^H(t) = (\pi_B^H(t), \pi_S^H(t))$ must be chosen so that $\pi_S^H(t) = \mathbf{0}$ and

$$\pi_B^H(t)^\top \sigma_B(t) = \frac{\gamma-1}{\gamma} \sigma_P(t, T)^\top.$$

This is satisfied if the hedge solely consists of the fraction $(\gamma-1)/\gamma$ of wealth invested in zero-coupon bonds maturing at time T . Hence, in this general k -factor Gaussian HJM model, a CRRA investor

can hedge all shifts in the opportunity set by investing in one bond only, namely the zero-coupon bond maturing at his investment horizon. Of course, any portfolio of term structure derivatives which has the same diffusion coefficients as the zero-coupon bond maturing at T provides an equally efficient hedge to changes in the term structure.

Assume without loss of generality that the zero-coupon bond maturing at T is the first of the k term-structure derivatives in (24) and let $\mathbf{e}_1$ be the d -dimensional vector $(1, 0, \dots, 0)^\top$. It follows from the portfolio decomposition (20) that the optimal investment strategy can be written as

$$\begin{pmatrix} \pi(t) \\ \pi_0(t) \end{pmatrix} = \begin{pmatrix} \frac{1}{\gamma} \pi_G(t) \\ 1 - \pi_G(t)^\top \mathbf{1} / \gamma \end{pmatrix} + \frac{\gamma - 1}{\gamma} \begin{pmatrix} \mathbf{e}_1 \\ -1 \end{pmatrix} = \frac{1}{\gamma} \begin{pmatrix} \pi_G(t) \\ 1 - \pi_G(t)^\top \mathbf{1} \end{pmatrix} + \frac{\gamma - 1}{\gamma} \begin{pmatrix} \mathbf{e}_1 \\ 0 \end{pmatrix}.$$

The optimal portfolio is a combination of the growth-optimal portfolio and the zero-coupon bond maturing at time T . We summarize our findings in the following proposition.

Proposition 1 *Assume a complete market with deterministic market prices of risk where the term structure dynamics is given by a k -factor Gaussian HJM-model. For any investor with CRRA utility of terminal wealth only and relative risk aversion coefficient γ , the optimal investment strategy is to invest a fraction $1/\gamma$ of wealth in the growth-optimal strategy and a fraction $(\gamma - 1)/\gamma$ of wealth in the zero-coupon bond maturing at the investment horizon.*

The higher the risk aversion coefficient, the higher the fraction of wealth in the zero-coupon bond and the lower the fraction of wealth in the growth-optimal portfolio. Investors less risk averse than logarithmic investors ($0 < \gamma < 1$) will short the zero-coupon bond, whereas more risk averse investors ($\gamma > 1$) will take a long position in the zero-coupon bond. Very risk-averse investors (γ approaching ∞) will invest almost their entire wealth in the zero-coupon bond and almost nothing in the growth-optimal strategy. For the original one-factor Vasicek (1977) model and constant market prices of risk, the same result was derived by Sørensen (1999). Note that other term structure derivatives enter the optimal investment strategy through the growth-optimal term and, hence, a complete set of $d + 1$ assets is needed to implement the strategy.

Merton (1992, Sec. 15.10) argues that in a market driven by m Markov state variables, the investors need access to (at least) $m + 2$ funds, namely (i) the savings account, (ii) the fund defined by the portfolio weights $\pi_G(t)/(\pi_G(t)^\top \mathbf{1})$ in the d risky assets and nothing in the savings account, and (iii) m funds hedging changes in the investment opportunity set. In the present non-Markov multi-factor setting, however, a single very simple hedge fund is sufficient. Moreover, Proposition 1 shows that all CRRA investors (or HARA investors, cf. the discussion below) having identical investment horizons will be satisfied with access to trade in just two funds, namely the growth-optimal fund and the zero-coupon bond maturing at the end of their investment horizon.

For the case in which the agent has HARA utility of the form $U_2(X, T) = (X - \underline{X})^{1-\gamma}/(1-\gamma)$, where $\underline{X}$ is some positive constant, we know from (23) that the optimal trading strategy is to invest $L(t_0) = E_{t_0}[\zeta_{t_0}(T)\underline{X}] = \underline{X}P(t_0, T)$ to cover the subsistence wealth level and invest the remaining $x - L(t_0)$ as in the CRRA case. From Proposition 1 we can therefore conclude that the optimal investment strategy is to invest $(x - \underline{X}P(t_0, T))/\gamma$ in the growth-optimal strategy and $(x - \underline{X}P(t_0, T))(\gamma - 1)/\gamma + \underline{X}P(t_0, T)$ in the zero-coupon bond maturing at time T . Therefore, a HARA investor will also invest in the growth-optimal strategy and the zero-coupon bond only so that the separation result still holds.

6 Optimal investment strategies with utility from consumption and terminal wealth

In this section we will find an explicit expression for the optimal investment strategy of an investor with utility both from intertemporal consumption and terminal wealth in the specialized market setting of Section 4. First, we focus on an investor with CRRA utility functions given by (13). In order to apply the general condition (21) we must find the dynamics of the process $Q(t)$ defined in (14). Applying (29) we can rewrite $Q(t)$ as

$$Q(t) = \int_t^T Z(t, s) ds + Z(t, T)$$

where

$$Z(t, s) = e^{-\frac{\beta}{\gamma}(s-t)} E_t \left[\zeta_t(s)^{\frac{\gamma-1}{\gamma}} \right] = \exp \left\{ -\frac{\beta}{\gamma}(s-t) + \frac{1-\gamma}{2\gamma^2} g(t, s) \right\} P(t, s)^{\frac{\gamma-1}{\gamma}}. \quad (33)$$

Itô's Lemma, (27), and (32) imply that

$$dZ(t, s) = Z(t, s) \left[\left(\frac{\beta}{\gamma} - \frac{1-\gamma}{2\gamma^2} \|\lambda(t)\|^2 + \frac{\gamma-1}{\gamma} r(t) + \frac{(1-\gamma)^2}{\gamma^2} \lambda_B(t)^\top \sigma_P(t, s) \right) dt + \frac{\gamma-1}{\gamma} \sigma_P(t, s)^\top dw_B(t) \right].$$

By the Leibnitz-type rule for stochastic processes stated in Lemma 1 in the Appendix we get that

$$\begin{aligned} dQ(t) &= \left[\left(\frac{\beta}{\gamma} + \frac{\gamma-1}{\gamma} r(t) - \frac{1-\gamma}{2\gamma^2} \|\lambda(t)\|^2 \right) Q(t) - 1 \right] dt \\ &\quad + \frac{(1-\gamma)^2}{\gamma^2} \lambda_B(t)^\top \left(\int_t^T Z(t, s) \sigma_P(t, s) ds + Z(t, T) \sigma_P(t, T) \right) dt \\ &\quad + \frac{\gamma-1}{\gamma} \left(\int_t^T Z(t, s) \sigma_P(t, s) ds + Z(t, T) \sigma_P(t, T) \right)^\top dw_B(t) \end{aligned}$$

corresponding to the diffusion coefficient

$$\sigma_Q(t)^\top = \left(\frac{\gamma-1}{\gamma} \frac{\left(\int_t^T Z(t, s) \sigma_P(t, s)^\top ds + Z(t, T) \sigma_P(t, T)^\top \right)^\top}{\int_t^T Z(t, s) ds + Z(t, T)}, \mathbf{0}^\top \right).$$

It follows from (21) that the hedge portfolio $\pi^H(t) = (\pi_B^H(t), \pi_S^H(t))$ must be chosen so that $\pi_S^H(t) = \mathbf{0}$ and

$$\pi_B^H(t)^\top \sigma_B(t) = \frac{\gamma - 1}{\gamma} \frac{\left(\int_t^T Z(t, s) \sigma_P(t, s)^\top ds + Z(t, T) \sigma_P(t, T)^\top \right)^\top}{\int_t^T Z(t, s) ds + Z(t, T)}.$$

A comparison with the volatility of a general continuous coupon bond given in (28) reveals that we can interpret the hedge portfolio $\pi_B^H(t)$ as an investment in a single bond with a continuous coupon of

$$k(s) = \frac{Z(t, s)}{P(t, s)} = \exp \left\{ -\frac{\beta}{\gamma}(s - t) + \frac{1 - \gamma}{2\gamma^2} g(t, s) \right\} P(t, s)^{-1/\gamma} \quad (34)$$

between time t and T and a lump sum payment at time T of

$$K = \frac{Z(t, T)}{P(t, T)} = \exp \left\{ -\frac{\beta}{\gamma}(T - t) + \frac{1 - \gamma}{2\gamma^2} g(t, T) \right\} P(t, T)^{-1/\gamma}. \quad (35)$$

We can give another characterization of the continuous-coupon hedge bond identified above using the forward martingale measures $\mathbb{Q}_s$ introduced by Jamshidian (1987) and Geman (1989). Expectations under the subjective probability measure $\mathbb{P}$, the spot martingale measure $\mathbb{Q}$, and the time s forward martingale measure $\mathbb{Q}_s$ satisfy the relation

$$\mathbb{E}_t[\zeta_t(s) X] = \mathbb{E}_t^\mathbb{Q} \left[e^{-\int_t^s r(u) du} X \right] = P(t, s) \mathbb{E}_t^{\mathbb{Q}_s} [X]$$

for any sufficiently well-behaved random variable X , cf., e.g., Duffie (1996, Ch. 6). Therefore, the coupon of the hedge bond can be rewritten using (33) and (15) as

$$k(s) = e^{-\frac{\beta}{\gamma}(s-t)} \mathbb{E}_t^{\mathbb{Q}_s} [\zeta_t(s)^{-1/\gamma}] = \frac{Q(t)}{X_{x, t_0}(t)} \mathbb{E}_t^{\mathbb{Q}_s} [c^*(s)] \quad (36)$$

and the terminal lump sum payment as

$$K = e^{-\frac{\beta}{\gamma}(T-t)} \mathbb{E}_t^{\mathbb{Q}_T} [\zeta_t(T)^{-1/\gamma}] = \frac{Q(t)}{X_{x, t_0}(t)} \mathbb{E}_t^{\mathbb{Q}_T} [X^*]. \quad (37)$$

The hedge applied at time t can now be seen as a bond that at each future date s pays a coupon at a rate proportional to the expectation under the forward martingale measure $\mathbb{Q}_s$ of the optimal consumption rate at time s and a terminal lump sum payment proportional to the expectation of the optimal terminal wealth under the forward martingale measure $\mathbb{Q}_T$. We summarize our findings in the following proposition.

Proposition 2 *Assume a complete market with deterministic market prices of risk where the term structure dynamics is given by a k -factor Gaussian HJM-model. For any investor with a CRRA utility from intermediate consumption and terminal time T wealth and relative risk coefficient γ , the optimal investment strategy is to invest a fraction $1/\gamma$ of wealth in the growth-optimal strategy and a fraction $(\gamma - 1)/\gamma$ of wealth in a bond maturing at T , paying a continuous coupon given by (36), and a lump sum payment given by (37) at T .*

For an investor with HARA utility functions given by (22), it follows from the proposition above and the discussion in Section 3 that the optimal strategy at time t_0 is to invest $(x - L(t_0))/\gamma$ in the growth-optimal strategy and the rest in bonds paying a total continuous coupon of $\underline{c}(s) + k(s)(\gamma - 1)/\gamma$ at each time s and a total lump sum payment of $\underline{X} + K(\gamma - 1)/\gamma$ at time T . Here $L(t_0) = \int_{t_0}^T \underline{c}(s)P(t_0, s) ds + \underline{X}P(t_0, T)$.

In Section 5 we demonstrated that all HARA investors having the same investment horizon and utility of terminal wealth only will agree upon the hedge portfolio. In the present case with intermediate consumption, this is no longer the case since the coupon rate and the lump-sum payment of the hedge bond depend on the intertemporal consumption pattern and, hence, on the risk aversion parameter γ , which can also be seen directly from (34) and (35).

7 Concluding remarks

In this paper we have derived optimal strategies for investments in stocks and term-structure derivatives for a CRRA or HARA investor in a complete market with multi-factor Gaussian HJM interest rates and deterministic market prices of risk. In particular, we have demonstrated how changes in the investment opportunity set can be hedged by a single bond: A zero-coupon bond for the case of utility from terminal wealth only and a continuous-coupon bond in the case of utility from intermediate consumption. Due to the limitations of the assumed dynamics of the financial markets, it is of course highly relevant to expand our study to models with more realistic term structure dynamics and other specifications of market prices of risk. We leave such an analysis to future research.

A Appendix: A Leibnitz-type rule for stochastic processes

Lemma 1 *Let $Z(t, s)$ be a family of stochastic variables such that for each fixed $s \in [t_0, T]$*

$$dZ(t, s) = \mu(t, s) dt + \sigma(t, s) dw(t), \quad t_0 \leq t \leq s,$$

where σ is sufficiently measurable and satisfies

$$(a) \int_{t_0}^T \sigma^2(t, s) dt < \infty \text{ for all } s \in [t_0, T],$$

$$(b) \int_{t_0}^T \left[\int_t^T \sigma(t, s) ds \right]^2 dt < \infty$$

almost surely. Then the dynamics of the stochastic process $Y(t) = \int_t^T Z(t, s) ds$ is given by

$$dY(t) = \left[\left(\int_t^T \mu(t, s) ds \right) - Z(t, t) \right] dt + \left(\int_t^T \sigma(t, s) ds \right) dw(t).$$

Proof: The proof is an application of the generalized Fubini-type rule for stochastic processes stated and applied in the Appendix of Heath, Jarrow, and Morton (1992). Since

$$Z(t_1, s) = Z(t_0, s) + \int_{t_0}^{t_1} \mu(t, s) dt + \int_{t_0}^{t_1} \sigma(t, s) dw(t),$$

we have

$$\begin{aligned} Y(t_1) &= \int_{t_1}^T Z(t_0, s) ds + \int_{t_1}^T \left[\int_{t_0}^{t_1} \mu(t, s) dt \right] ds + \int_{t_1}^T \left[\int_{t_0}^{t_1} \sigma(t, s) dw(t) \right] ds \\ &= \int_{t_1}^T Z(t_0, s) ds + \int_{t_0}^{t_1} \left[\int_{t_1}^T \mu(t, s) ds \right] dt + \int_{t_0}^{t_1} \left[\int_{t_1}^T \sigma(t, s) ds \right] dw(t) \\ &= Y(t_0) + \int_{t_0}^{t_1} \left[\int_t^T \mu(t, s) ds \right] dt + \int_{t_0}^{t_1} \left[\int_t^T \sigma(t, s) ds \right] dw(t) \\ &\quad - \int_{t_0}^{t_1} Z(t_0, s) ds - \int_{t_0}^{t_1} \left[\int_t^{t_1} \mu(t, s) ds \right] dt - \int_{t_0}^{t_1} \left[\int_t^{t_1} \sigma(t, s) ds \right] dw(t) \\ &= Y(t_0) + \int_{t_0}^{t_1} \left[\int_t^T \mu(t, s) ds \right] dt + \int_{t_0}^{t_1} \left[\int_t^T \sigma(t, s) ds \right] dw(t) \\ &\quad - \int_{t_0}^{t_1} Z(t_0, s) ds - \int_{t_0}^{t_1} \left[\int_{t_0}^s \mu(t, s) dt \right] ds - \int_{t_0}^{t_1} \left[\int_{t_0}^s \sigma(t, s) dw(t) \right] ds \\ &= Y(t_0) + \int_{t_0}^{t_1} \left[\int_t^T \mu(t, s) ds \right] dt + \int_{t_0}^{t_1} \left[\int_t^T \sigma(t, s) ds \right] dw(t) - \int_{t_0}^{t_1} Z(t, t) dt \end{aligned}$$

where the Fubini rule was used in the second and fourth equality. The claim now follows. $\square$

References

- Amin, K. I. and R. A. Jarrow (1992). Pricing Options on Risky Assets in a Stochastic Interest Rate Economy. *Mathematical Finance* 2(4), 217–237.
- Bardhan, I. (1994). Consumption and Investment under Constraints. *Journal of Economic Dynamics and Control* 18, 909–929.
- Brace, A. and M. Musiela (1994). A Multifactor Gauss Markov Implementation of Heath, Jarrow, and Morton. *Mathematical Finance* 4(3), 259–283.
- Brennan, M. J., E. S. Schwartz, and R. Lagnado (1997). Strategic Asset Allocation. *Journal of Economic Dynamics and Control* 21(8–9), 1377–1403.
- Carverhill, A. (1994). When is the Short Rate Markovian? *Mathematical Finance* 4(4), 305–312.
- Chacko, G. and L. M. Viceira (1999, June). Dynamic Consumption and Portfolio Choice with Stochastic Volatility in Incomplete Markets. Working paper, Harvard University.
- Cox, J. C. and C.-f. Huang (1989). Optimal Consumption and Portfolio Policies when Asset Prices Follow a Diffusion Process. *Journal of Economic Theory* 49, 33–83.

- Cox, J. C. and C.-f. Huang (1991). A Variational Problem Arising in Financial Economics. *Journal of Mathematical Economics* 20, 465–487.
- Cuoco, D. (1997). Optimal Consumption and Equilibrium Prices with Portfolio Constraints and Stochastic Income. *Journal of Economic Theory* 71(1), 33–73.
- Cvitanic, J. and I. Karatzas (1992). Convex Duality in Constrained Portfolio Optimization. *The Annals of Applied Probability* 2(4), 767–818.
- Detemple, J., R. Garcia, and M. Rindisbacher (1999, June). A Monte-Carlo Method for Optimal Portfolios. Working Paper.
- Duffie, D. (1996). *Dynamic Asset Pricing Theory* (Second ed.). Princeton, New Jersey, USA: Princeton University Press.
- Duffie, D., W. Fleming, H. M. Soner, and T. Zariphopoulou (1997). Hedging in Incomplete Markets with HARA Utility. *Journal of Economic Dynamics and Control* 21(4–5), 753–782.
- Geman, H. (1989). The Importance of the Forward Neutral Probability in a Stochastic Approach of Interest Rates. Working paper, ESSEC.
- Heath, D., R. Jarrow, and A. Morton (1992). Bond Pricing and the Term Structure of Interest Rates: A New Methodology for Contingent Claims Valuation. *Econometrica* 60(1), 77–105.
- Jamshidian, F. (1987). Pricing of Contingent Claims in the One Factor Term Structure Model. Working paper, Merrill Lynch Capital Markets.
- Karatzas, I., J. P. Lehoczky, and S. E. Shreve (1987). Optimal Portfolio and Consumption Decisions for a “Small Investor” on a Finite Horizon. *SIAM Journal on Control and Optimization* 25(6), 1557–1586.
- Karatzas, I. and S. E. Shreve (1998). *Methods of Mathematical Finance*, Volume 39 of *Applications of Mathematics*. New York: Springer-Verlag.
- Kim, T. S. and E. Omberg (1996). Dynamic Nonmyopic Portfolio Behavior. *The Review of Financial Studies* 9(1), 141–161.
- Korn, R. (1997). *Optimal Portfolios*. World Scientific.
- Merton, R. C. (1969). Lifetime Portfolio Selection Under Uncertainty: The Continuous-Time Case. *Review of Economics and Statistics* 51, 247–257. Reprinted as Chapter 4 in Merton (1992).
- Merton, R. C. (1971). Optimum Consumption and Portfolio Rules in a Continuous-Time Model. *Journal of Economic Theory* 3, 373–413. Erratum: Merton (1973a). Reprinted as Chapter 5 in Merton (1992).

- Merton, R. C. (1973a). Erratum. *Journal of Economic Theory* 6, 213–214.
- Merton, R. C. (1973b). An Intertemporal Capital Asset Pricing Model. *Econometrica* 41(5), 867–887. Reprinted in an extended form as Chapter 15 in Merton (1992).
- Merton, R. C. (1992). *Continuous-Time Finance*. Padstow, UK: Basil Blackwell Inc.
- Munk, C. (1999). Optimal Consumption/Investment Policies with Undiversifiable Income Risk and Liquidity Constraints. Working paper, Odense University. Forthcoming in the *Journal of Economic Dynamics and Control*.
- Ocone, D. L. and I. Karatzas (1991). A Generalized Clark Representation Formula, with Application to Optimal Portfolios. *Stochastics and Stochastic Reports* 34, 187–220.
- Sørensen, C. (1999). Dynamic Asset Allocation and Fixed Income Management. Working paper, Copenhagen Business School. Forthcoming in *Journal of Financial and Quantitative Analysis*.
- Vasicek, O. (1977). An Equilibrium Characterization of the Term Structure. *Journal of Financial Economics* 5, 177–188.